

Tabulation Method (or) Quine-McClusky Method

Tabulation Method

Problem with K-Map

- Convenient if number of variables is less
- Trial-and-error procedure that relies on human ability to recognize certain patterns

Tabulation Method

- Can be applied to problems with any number of variables
- Suitable for machine computation

(a)

	w	x	y	z	
0	0	0	0	0	✓
1	0	0	0	1	✓
2	0	0	1	0	✓
8	1	0	0	0	✓
10	1	0	1	0	✓
11	1	0	1	1	✓
14	1	1	1	0	✓
15	1	1	1	1	✓

Simplify the following Boolean function by using the tabulation method:

$$F = \Sigma(0, 1, 2, 8, 10, 11, 14, 15)$$

Step 1: Group binary representation of the minterms according to the number of 1's contained, as shown in Table 3-5, column (a). This is done by grouping the minterms into five sections separated by horizontal lines. The first section contains the number with no 1's in it. The second section contains those numbers that have only one 1. The third, fourth, and fifth sections contain those binary numbers with two, three, and four 1's, respectively. The decimal equivalents of the minterms are also carried along for identification.

identification.

Step 2: Any two minterms that differ from each other by only one variable can be combined, and the unmatched variable removed. Two minterm numbers fit into this category if they both have the same bit value in all positions except one. The minterms in one section are compared with those of the next section down only, because two terms differing by more than one bit cannot match. The minterm in the first section is compared with each of the three minterms in the second section. If any two numbers are the same in every position but one, a check is placed to the right of both minterms to show that they have been used. The resulting term, together with the decimal equivalents, is listed in column (b) of the table. The variable eliminated during the matching is denoted by a dash in its original position. In this case, m_0 (0000) combines with m_1 (0001) to form (000-). This combination is equivalent to the algebraic operation

$$m_0 + m_1 = w'x'y'z' + w'x'y'z = w'x'y'$$

Minterm m_0 also combines with m_2 to form (00-0) and with m_8 to form (-000). The result of this comparison is entered into the first section of column (b). The minterms of sections two and three of column (a) are next compared to produce the terms listed in the second section of column (b). All other sections of (a) are similarly compared and subsequent sections formed in (b). This exhaustive comparing process results in the four sections of (b).

(a)	(b)
$w x y z$	$w x y z$
0 0 0 0 0 ✓	0, 1 0 0 0 -
	0, 2 0 0 - 0 ✓
	0, 8 - 0 0 0 ✓
1 0 0 0 1 ✓	
2 0 0 1 0 ✓	2, 10 - 0 1 0 ✓
8 1 0 0 0 ✓	8, 10 1 0 - 0 ✓
10 1 0 1 0 ✓	10, 11 1 0 1 - ✓
	10, 14 1 - 1 - ✓
11 1 0 1 1 ✓	
14 1 1 1 0 ✓	11, 15 1 - 1 1 ✓
	14, 15 1 1 1 - ✓
15 1 1 1 1 ✓	

(a)	(b)	(c)
$w x y z$	$w x y z$	$w x y z$
0 0 0 0 ✓	0, 1 0 0 0 - 0, 2 0 0 - 0 ✓	0, 2, 8, 10 - 0 - 0 0, 8, 2, 10 - 0 - 0
1 0 0 0 1 ✓ 2 0 0 1 0 ✓	0, 8 - 0 0 0 ✓	10, 11, 14, 15 1 - 1 - 10, 14, 11, 15 1 - 1 -
8 1 0 0 0 ✓	2, 10 - 0 1 0 ✓ 8, 10 1 0 - 0 ✓	
10 1 0 1 0 ✓	10, 11 1 0 1 - ✓	
11 1 0 1 1 ✓ 14 1 1 1 0 ✓	10, 14 1 - 1 - ✓	
15 1 1 1 1 ✓	11, 15 1 - 1 1 ✓ 14, 15 1 1 1 - ✓	

Step 3: The terms of column (b) have only three variables. A 1 under the variable means it is unprimed, a 0 means it is primed, and a dash means the variable is not included in the term. The searching and comparing process is repeated for the terms in column (b) to form the two-variable terms of column (c). Again, terms in each section need to be compared only if they have dashes in the same position. Note that the term (000-) does not match with any other term. Therefore, it has no check mark at its right. The decimal equivalents are written on the left-hand side of each entry for identification purposes. The comparing process should be carried out again in column (c) and in subsequent columns as long as proper matching is encountered. In the present example, the operation stops at the third column.

(a)	(b)	(c)
$w x y z$	$w x y z$	$w x y z$
0 0 0 0 ✓	0, 1 0 0 0 -	0, 2, 8, 10 - 0 - 0
	0, 2 0 0 - 0 ✓	0, 8, 2, 10 - 0 - 0
1 0 0 0 1 ✓	0, 8 - 0 0 0 ✓	10, 11, 14, 15 1 - 1 -
2 0 0 1 0 ✓		10, 14, 11, 15 1 - 1 -
8 1 0 0 0 ✓	2, 10 - 0 1 0 ✓	
	8, 10 1 0 - 0 ✓	
10 1 0 1 0 ✓	10, 11 1 0 1 - ✓	
11 1 0 1 1 ✓	10, 14 1 - 1 - ✓	
14 1 1 1 0 ✓		
	11, 15 1 - 1 1 ✓	
15 1 1 1 1 ✓	14, 15 1 1 1 - ✓	

Step 4: The unchecked terms in the table form the prime implicants. In this example, we have the term $w'x'y'$ (000-) in column (b), and the terms $x'z'$ (-0-0) and wy (1-1-) in column (c). Note that each term in column (c) appears twice in the table, and as long as the term forms a prime implicant, it is unnecessary to use the same term twice. The sum of the prime implicants gives a simplified expression for the function.

$$F = w'x'y' + x'z' + wy$$